A direct count comparison over the full acquired probability range

Research note for author review

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1 One small-noise comparison for all acquired probabilities

The direct geometric localization leaves probabilities between 1/4 and 53/100. The left panels need a slightly wider conditional support strip than the right panels. This does not require a different count argument. The following theorem uses the common amplitude 7/10. It retains the third-moment correction, which can have the unfavorable sign on the left.

Theorem 1. *Let $n \geq 42$, $1/4 \leq p \leq 53/100$, $q = 1 - p$, and $Y = B + U$, where $\mathbb{P}(B = 1) = p$ and $\mathbb{E}(U \mid B) = 0$. Suppose the two conditional laws obey*

$$|U_j| \leq A := 7/10, \qquad U_0 \leq a_0, \quad U_1 \geq -a_1, \qquad a_0, a_1 \geq 0, \quad a_0 + a_1 \leq d_* := 17/250.$$

Put $v = pq$, $s = \mathbb{E}U^2$, $\lambda = ns$, $\ell = (v + s)^{-1/2}$, $X = \ell(Y - p)$ and $\rho = \mathbb{E}|X|^3$. Let S_n be the sum of n independent copies of X. Assume

$$0 < \lambda \leq U_* := 1/500, \qquad |H| \leq 46/25.$$

Choose an integer k so that $np + H/\ell = k + u$ with

$$-109/500 \leq u \leq 391/500.$$

Such an anchor always exists, and $1 \leq k \leq n - 1$. Write $\delta = k - np$, $\rho_B = (1 - 2v)/\sqrt{v}$ and let F_B denote the distribution function of $\mathrm{Bin}(n, p)$. *For every $0 \leq C \leq 41/100$,*

$$\begin{aligned}\Delta_C = \sqrt{n}\{\mathbb{P}(S_n \leq H) - \Phi(H/\sqrt{n})\} - C\rho \\ - \sqrt{n}\{F_B(k) - \Phi(\delta/\sqrt{nv})\} + C\rho_B < -\lambda/100.\end{aligned}$$

For $C = C_E = (3 + \sqrt{10})/(6\sqrt{2\pi}) < 41/100$, the external strict Bernoulli theorem excludes a nonnegative fixed-target excess. At $\lambda = 0$ the law is Bernoulli and that theorem applies directly. The positive-noise statement has no contact or predecessor premise. Such premises remain in the separate acquisition of its strip and threshold bounds.

1.1 Exact conditioning and the finite normalization

Set $l = -109/500$, $h = 391/500$, $b = 1/20$, $v_* = 3/16$ and $D = 171/100$. The phase interval has width one and equals $[(A-1-2d_*)/2, (A+1-2d_*)/2]$. In particular choosing $k = \lfloor np+H/\ell-l \rfloor$ is always possible. Let W_j be a sum of j independent variables with law U_1 and $n-j$ with law U_0. Write $\pi_j = \binom{n}{j} p^j q^{n-j}$, zero outside the legal counts, and define

$$f(u) = \frac{\Phi((\delta + u)/\sqrt{n(v + s)}) - \Phi(\delta/\sqrt{n(v + s)})}{\pi_k},$$

$$O = \mathbb{P}(W_k > u) + f(u), \qquad T = \sum_{i=1}^{n-k} \mathbb{P}(W_{k+i} \leq u - i).$$

Conditioning on the actual Bernoulli count gives

$$\begin{aligned}\mathbb{P}(S_n \le H) - F_B(k) = &-\sum_{i=0}^{k} \pi_{k-i}\mathbb{P}(W_{k-i} > u+i)\\ &+\sum_{i=1}^{n-k} \pi_{k+i}\mathbb{P}(W_{k+i} \le u-i).\end{aligned}$$

The outgoing event is strict and the incoming event closed, including at atoms. If $\pi_{k+i}/\pi_k \le R$, discarding all but the own outgoing count gives

$$\Delta_C \le -\sqrt{n}\pi_k(O - RT) + \mathcal{R}_C, \tag{1}$$

where

$$\mathcal{R}_C = \sqrt{n}\{\Phi(\delta/\sqrt{nv}) - \Phi(\delta/\sqrt{n(v+s)})\} + C(\rho_B - \rho).$$

We have $v \ge v_*$ and $\sqrt{v+s} < 501/1000$. Therefore $|\delta| < D$, and $|\delta| < 1$ if $|u| \le b$. The segment from p to k/n lies inside $(1/5, 29/50)$; this also proves that the count is interior. On that segment $r(1-r) \ge 4/25$ and $|1-2r| \le 3/5$, so

$$|D'''(r\|p)| < 24, \qquad |(\log(r(1-r)))'| < 4, \qquad \frac{1}{12r(1-r)} < \frac{53}{100}.$$

Finite Stirling and Taylor's theorem yield

$$\sqrt{n}\pi_k = \frac{\phi(0)}{\sqrt{v}} e^{-\delta^2/(2nv)+e}, \qquad n|e| < Z(D_0) := 2D_0 + 53/100 + 4D_0^3/42 \tag{2}$$

when $|\delta| \le D_0 \le D$. The finite factorial formula is

$$\log(j!) = (j+1/2)\log j - j + \tfrac{1}{2}\log(2\pi) + r_j, \qquad 0 < r_j < 1/(12j).$$

Its proof is retained in the manuscript's count appendix, beginning at `appendices/count-comparison.tex:131`. The three errors in (2) are at most $4D_0^3/n^2$, $2D_0/n$ and $53/(100n)$. Absolute derivatives are used; the segment may cross $1/2$.

Holding the actual count fixed gives

$$\log f'(t) = -\tfrac{1}{2}\log(1+s/v) + \frac{\delta^2}{2nv} - \frac{(\delta+t)^2}{2n(v+s)} - e.$$

Since $1/(2v_*) + D^2/(2\cdot 42 v_*^2) < 4$, for $|t| \le b_0$ this implies

$$n|\log f'(t)| \le 4\lambda + \frac{2D_0 b_0 + b_0^2}{2v_*} + Z(D_0).$$

Use $(D_0, b_0) = (1, b)$ or (D, h), together with $e^{-x} \ge 1-x$, $e^x \le (1-x)^{-1}$ for $x < 1$, and $0.39886 < \phi(0) < 0.4$. The resulting uniform bounds are

	$\lvert u\rvert \le 1/20$	$l \le u \le h$
$\sqrt{n}\pi_k$ lower bound	69/100	14/25
f' lower bound	9/10	2/3
f' upper bound	27/25	3/2
π_{k+i}/π_k upper bound	107/100	9/5
$\mathcal{R}_C$ upper bound	$4\lambda/n$	$6\lambda/n$

At a binomial mode the displacement is at most one, and (2) gives $\sqrt{n}\max_j \pi_j < 1$. Dividing by 14/25 proves the global ratio bound. In the inner range the first exact ratio is at most $1+(53/100)/(42v_*) < 107/100$; subsequent ratios are at most one since $(n+1)p - k < 153/100 < 2$.

1.2 The third-moment correction on the left

We justify the repair row without assuming $\rho \ge \rho_B$. The inward bounds fix both signs because $a_0 < p$ and $a_1 < q$. The conditional cubic expansion gives

$$\mathbb{E}|Y-p|^3 \ge v(1-2v) + (3\min(p,q) - d_*)s.$$

Indeed $U_0^3 \le a_0 U_0^2$ and $U_1^3 \ge -a_1 U_1^2$. Using $(1+x)^{3/2} \le 1 + 3x/2 + 3x^2/8$ for $x \ge 0$, we obtain

$$(v+s)^{3/2}(\rho - \rho_B) \ge s\{3\min(p,q) - \tfrac{3}{2}(1-2v) - d_* - 3(1-2v)s/(8v)\} \ge -B_* s,$$

where

$$B_* = \tfrac{3}{2}(1-2v_*) + d_* - \tfrac{3}{4} + \frac{3(1-2v_*)U_*}{8 \cdot 42 v_*}.$$

An exact square comparison proves $(41/100)B_*/v_*^{3/2} < 4/3$. Consequently $C(\rho_B - \rho) \le (4/3)s$ for every $0 \le C \le 41/100$. The normal variance correction is at most $(5/2)\delta_+ s$, since $0.2/v_*^{3/2} < 5/2$. Hence

$$\mathcal{R}_C \le \{(5/2)\delta_+ + 4/3\}\lambda/n,$$

which gives both repair constants in the table.

The sign issue is real. At $p = 13/50$, take $U_1 = 0$ and $U_0 = \pm 1/1000$ with equal conditional probabilities. Then $s = q/10^6$ and $\mathbb{E}|Y-p|^3 = v(1-2v) + 3ps$. Direct rational squaring gives $\rho < \rho_B$. This law obeys the present support and noise restrictions; it refutes discarding the correction, not the theorem.

1.3 An own-count bound and a complete incoming tail

If $s_j = \mathbb{E}U_j^2$, then $|s_1 - s_0| \le 4s$ and $\mathbb{E}W_j^2 = \lambda + (j - np)(s_1 - s_0)$. Thus

$$\frac{19}{21}\lambda \le t := \mathbb{E}W_k^2 \le \frac{23}{21}\lambda \quad (|u| \le b), \qquad t \le \frac{7}{6}\lambda \quad \text{globally.}$$

For incoming counts, $\mathbb{E}W_{k+i}^2 \le iV\lambda$, with $V = 6/5$ in the inner range and $V = 4/3$ globally.

For a centered independent sum of variance t and summands bounded in absolute value by A, independence gives $|\mathbb{E}W^3| \le At$ and $\mathbb{E}W^4 \le A^2 t + 3t^2$. Set $\alpha = 3/5$ and

$$h_\alpha(x) = \frac{-x^4 + 3\alpha^2 x^2 + 2\alpha^3 x}{5\alpha^4} \le \mathbf{1}_{(0,\infty)}(x).$$

The numerator factors as $-x(x+\alpha)^2(x-2\alpha)$; its positive maximum is $(9+6\sqrt{3})\alpha^4/4 < 5\alpha^4$. After shifting by σr, adding br, dividing by $t > 0$ and setting $r = zt$, the resulting lower bound is

$$F_{\sigma,b}(z,t) = \frac{3\alpha^2 - A^2 - 3t}{5\alpha^4} + \left(\frac{2\sigma}{5\alpha} + b - \frac{4At}{5\alpha^4}\right) z + \frac{3\alpha^2 t - 6t^2}{5\alpha^4} z^2 - \frac{t^3 z^4}{5\alpha^4}.$$

It is concave in $t \ge 0$. On $0 \le z \le z_0$, replace z^4 by $z_0^2 z^2$. The resulting lower quadratic is checked at its endpoints and eligible vertex, for $t = 0$ and $t = T_* := 23/10500$. This proves

$$F_{1,-27/25} > 1/4 \quad (0 \le z \le 3/2), \qquad F_{-1,9/10} > 4/5 \quad (0 \le z \le 1).$$

The exact margins are at least $10788447499/444528000000$ and $569/5670$. The edge $t = 0$ belongs to the continuous divided polynomial; the actual variance is positive because $\lambda > 0$.

For negative phase $u = -r$, after $r = 3t/2$ use Cantelli:

$$O \geq \frac{r^2}{t + r^2} - \frac{27}{25} r.$$

At the switch its value exceeds $t/4$ and its derivative is positive. The derivative first increases and then decreases, so any later zero is a maximum. The other endpoint $r = b$ also exceeds $t/4$, as checked at $t = T_*$. Both switches lie below b. For positive phase after $r = t$, use $O \geq f(r) \geq (9/10)r > (4/5)t$. We conclude that

$$O > \frac{1}{4}\frac{19}{21}\lambda \qquad (|u| \leq b). \tag{3}$$

For incoming counts use the quadratic-hinge Poisson comparison of Pinelis, *On the Bennett–Hoeffding inequality*, arXiv:0902.4058, Proposition 2.8 and Corollary 2.11. Independent centered summands bounded above by A and having total variance at most V_0 satisfy

$$\mathbb{P}(Z \geq x) \leq \frac{\mathbb{E}(P_\mu - j)_+^2}{(x/A + \mu - j)^2}, \qquad \mu = V_0/A^2, \quad j < x/A + \mu.$$

This follows by applying its moment comparison to the quadratic hinge at $A(j - \mu)$ and then Markov. The variance upper bound and closed event are part of this application. Apply it to $-W_{k+i}$.

For $i = 1$ and $u \leq h_0 < 1 - A$, choose $j = 1$ to obtain

$$Q(\mu, h_0) = \frac{\mu^2(1/2 + \mu/6)}{((1 - h_0)/A - 1 + \mu)^2}, \qquad \mu = V\lambda/A^2.$$

Here $\mathbb{E}(P_\mu - 1)_+^2 = \mu^2 - \mu + 1 - e^{-\mu} \leq \mu^2/2 + \mu^3/6$ by Taylor's theorem. For integer $i \geq 2$, the elementary inequality $(P - i)_+^2 \leq \binom{P}{i+1}$ gives numerator at most $(i\mu)^{i+1}/(i+1)!$ after variance inflation to $iV\lambda$. For $i = 2$, the pointwise assertion reduces to $(r - 1)(r - 2) \geq 0$ for integer $r = P - i \geq 1$; increasing i increases the binomial coefficient. The expectation follows from the Poisson series.

For a starting count $m = 2$ or 3, put

$$b_m(h_0) = 1/A - 1 - h_0/(mA), \qquad J_2(\mu, h_0) = \frac{\mu^3}{3(b_2(h_0) + \mu)^2(1 - 3\mu)},$$

$$J_3(\mu, h_0) = \frac{3\mu^4}{8(b_3(h_0) + \mu)^2(1 - 3\mu)}.$$

If $h_0 \geq 0$, $b_m(h_0) > 0$ and $3\mu < 1$, these bound the entire incoming sum from $i = m$ onwards. Indeed the hinge $j = i$ has denominator at least $i(b_m(h_0) + \mu)$, and the ratio of the coefficients $i^{i-1}/(i+1)!$ is $[i/(i+2)](1 + 1/i)^i < e < 3$. This proves the displayed geometric sums without truncating the tail. At the largest phase h, $b_2(h) < 0$, so we use J_3 there and bound the first two terms separately. In contrast $b_3(h) = 59/1050 > 0$.

1.4 Three closed phase intervals

All constants below are evaluated at $U_* = 1/500$. Set

$$\mu_I = (6/5)U_*/A^2, \qquad \mu_G = (4/3)U_*/A^2.$$

For the inner phase, $Q(\mu, b)/\lambda$ and $J_2(\mu, b)/\lambda$ increase with μ on the stated range. For Q this follows from $\mu \leq (1 - b)/A - 1$; for J_2 it follows by factoring it into $[\mu/(b_2 + \mu)]^2/(1 - 3\mu)$ times

a positive constant. Therefore, with $T_I = Q(\mu_I, b) + J_2(\mu_I, b)$, the comparison (1) and (3) give the per-λ margin

$$\frac{69}{100}\left\{\frac{1}{4}\frac{19}{21} - \frac{107}{100}\frac{T_I}{U_*}\right\} - \frac{4}{42} - \frac{1}{100} = \frac{17926467866165398367}{1060536251855551002000} > 0.$$

This is uniform down to arbitrarily small positive noise.

For $l \le u \le -b$, set $T_G = 7U_*/6$ and

$$L_- := \min_{r\in\{b,-l\}}\left\{\frac{r^2}{T_G + r^2} - \frac{3}{2}r\right\}.$$

The expression is concave on the whole interval because $3b^2 > T_G$. Thus $O \ge L_-$ by Cantelli, and $T \le Q(\mu_G, 0) + J_2(\mu_G, 0)$. The final margin at the upper noise endpoint is

$$\frac{14}{25}\{L_- - \tfrac{9}{5}[Q(\mu_G, 0) + J_2(\mu_G, 0)]\} - \frac{6U_*}{42} - \frac{U_*}{100} = \frac{303207159039}{1226220050000} > 0.$$

For $b \le u \le h$, set $t_G = 4U_*/3$ and define

$$L_+ := \min_{u\in\{b,h\}}\left\{\frac{2}{3}u - \frac{9}{5}\left[\frac{t_G}{t_G + (1-u)^2} + \frac{2t_G}{2t_G + (2-u)^2}\right]\right\}.$$

The expression minimized is concave on the entire phase interval, since $3(1-h)^2 > t_G$ and $3(2-h)^2 > 2t_G$. Use the lower normal increment, Cantelli for the first two incoming counts, and J_3 for the rest. The remaining margin is

$$\frac{14}{25}\{L_+ - \tfrac{9}{5}J_3(\mu_G, h)\} - \frac{6U_*}{42} - \frac{U_*}{100} = \frac{1429750577643511489}{102275398586900550000} > 0.$$

All three brackets are positive before the lower own-count mass is used. For the two outer intervals, replacing actual λ by U_* only decreases the chosen own-count bound and increases tails and repairs. The three closed intervals cover $[l, h]$ including their common endpoints. This proves Theorem 1.

Scope of the reduction. Combining the new left-panel 0.7 strips with the previously acquired right-panel 0.6 strips supplies the common amplitude used here. The same-law conversion is $U = (X - \mathbb{E}[X \mid \mathbf{1}_{X>0}])/\ell$, $s = \theta/\ell^2$ and $\lambda = ns$; the inward displacements sum to the geometric $d < .068$. The coarse raw-threshold bound is below 1.84. This gives one small-noise exclusion on the entire acquired probability range. The larger-noise comparison and the independent finite base remain open. No older certificate or manuscript source is removed by this note.